

Actuarial Corporate Finance

Lesson 5

Security Market Line

Prof. Prabhdeep Singh

Administrative

- ▶ Everyone should have a partner and time slot for their presentation by now. Topics for a few people still pending finalization.
- ▶ Rubric for presentation (100pts):
 - Meeting 1 (10 pts): Gather materials and discuss initial thoughts with me.
 - Meeting 2 (10 pts): Show draft slides and discuss with me what you are going to say on each slide.
 - Presentation content (50 pts): At the depth appropriate for the course.
 - Presentation clarity (15 pts): The slides and what you say on each slide is clear.
 - Presentation delivery (15 pts): Material and how it is presented is engaging for the audience.
 - AI Policy for Presentation only: If you decide to use AI to enhance slides, then you must submit three sets of slides to me: (1) Slides without enhancement (2) Slides with AI enhancement (3) Final slides with your edits to (2) You can **NOT** use AI to generate script for your slides. These must be your own words.

Review of Lesson 4 - CAPM

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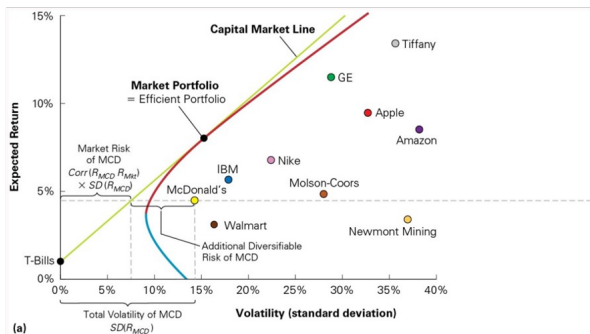


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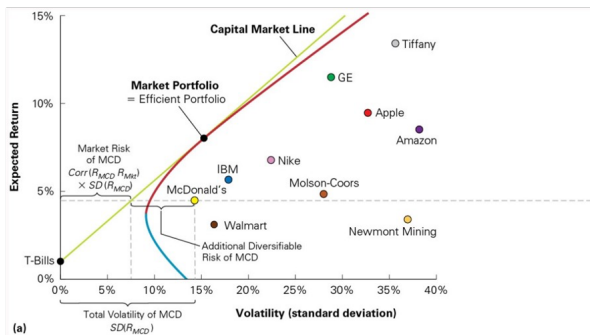
► For any investment i : $\mathbb{E}[R_i] = r_f + \rho_{i,M} \cdot \sigma_i \cdot \left(\frac{\mathbb{E}[R_M] - r_f}{\sigma_M} \right)$

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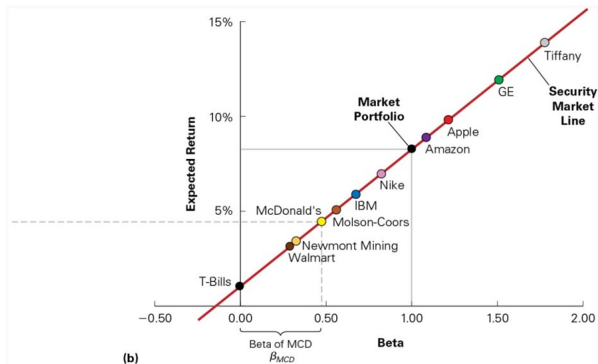
► Exercise W 5-1

Review of Homework

Security Market Line

► **Security Market Line (SML):** graph of expected return against β

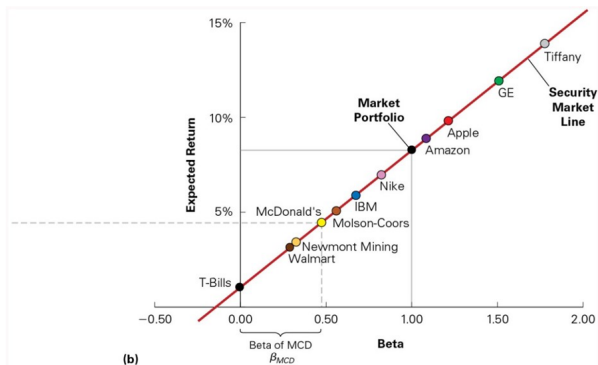
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► **Slope of SML:** market risk premium $\mathbb{E}[R_M] - r_f$

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- ▶ *Exercise W 5-2*

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- False. $\rightarrow$ Returns from distant past may not be representative of firm's current strategy & current economic conditions.

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 - No (sort of). $\rightarrow$ Share price is cut in half; I double the number of shares of BoA stock, but don't need to spend any money for that, and I maintain the same proportion of wealth invested in BoA.
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- Yes. → There will be 50% more shares of *Walmart* stock, so my PF should also have 50% more shares. Since firm is raising new funds, *Walmart's* market share will increase by 50%, stock price remains (roughly) same. Thus I should be investing a higher portion of my wealth into *Walmart*, at the expense of other stocks.

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- ▶ Next: how to estimate CAPM parameters?